Basics

Start and echo

Script Start

`@echo off`	hide command lines
`echo Hello`	print text
`echo.`	blank line
`REM comment`	comment
`:: comment`	quick comment
`pause`	wait for key
`cls`	clear screen
`exit /b 0`	end script
`title My App`	window title
`color 0A`	green on black

Run

`script.bat`	run a batch file
`call other.bat`	run and return
`start "" notepad`	open program
`cmd /c command`	run and close

Echo & Output

Show text on screen

Echo Basics

`echo Hello World`	print a line
`echo.`	print empty line
`echo %%`	print one percent sign
`echo ^& ^| ^> ^<`	escape special chars
`echo ^^`	print a caret
`echo off`	hide commands (no @)
`echo`	show echo status

Colors & Window

`color 0A`	green on black
`color 1F`	white on blue
`color 4E`	yellow on red
`color`	reset colors
`title My Script`	set window title
`mode con cols=80 lines=25`	set window size

Screen Tricks

`cls`	clear the screen
`echo Done! & pause`	print then wait
`pause >nul`	wait, no message
`timeout /t 3 /nobreak`	wait, ignore keys
`echo ^G`	beep sound

Variables

Set, math, input

Variables

`set name=Ali`	set variable
`echo %name%`	use variable
`set /p name=Name:`	read input
`set /a n=5+3`	do math
`set /a n+=1`	increment
`set name=`	delete variable

Special Variables

`%1  %2  %*`	arguments
`%~dp0`	script folder
`%cd%`	current folder
`%date%  %time%`	date / time
`%random%`	random number
`%errorlevel%`	last exit code
`%username%  %computername%`	user / pc name

Strings

`%name:~0,3%`	first 3 chars
`%name:old=new%`	replace text
`setlocal enabledelayedexpansion`	!var! in loops

Math & Strings

Numbers and text tricks

Math with set /a

`set /a x=5+3`	add
`set /a x=10-4`	subtract
`set /a x=6*7`	multiply
`set /a x=20/3`	whole division (6)
`set /a x=20%%3`	remainder (in file)
`set /a x+=2`	add to variable
`set /a x=(2+3)*4`	use parentheses
`set /a x=%random% %% 100`	random 0-99
`set /a "x=5 & 3"`	bitwise AND

Substrings

`%s:~0,3%`	first 3 characters
`%s:~-3%`	last 3 characters
`%s:~2%`	skip first 2
`%s:~1,-1%`	trim both ends

Replace & Test

`%s:a=b%`	replace a with b
`%s: =%`	remove all spaces
`if "%s%"=="" echo empty`	check empty text
`if /i "%s%"=="YES"`	ignore case compare

Arguments & Input

Talk to the user

Script Arguments

`%0`	script name
`%1 %2 %3`	first three arguments
`%*`	all arguments
`%~1`	argument without quotes
`shift`	move arguments left
`if "%~1"=="" goto usage`	no argument given

Path Modifiers

`%~dp0`	script drive + folder
`%~n0`	script name only
`%~x1`	extension of arg 1
`%~f1`	full path of arg 1
`%~z1`	file size of arg 1

User Input

`set /p name=Your name:`	ask a question
`set /p ok=Continue (Y/N)?`	ask yes or no
`choice /c YN /m "Sure?"`	one key choice
`if errorlevel 2 goto no`	choice N was pressed
`set /p x=<file.txt`	read first line of file

Flow Control

If, for, goto

If

`if "%a%"=="1" echo yes`	text equal
`if %n% equ 5`	number equal
`neq lss leq gtr geq`	number compare
`if not exist file.txt`	file missing?
`if defined name`	variable set?
`if ... ( ... ) else ( ... )`	else block
`if errorlevel 1`	last command failed

Loops

`for %%i in (1 2 3) do echo %%i`	loop items
`for /l %%i in (1,1,5) do echo %%i`	count loop
`for %%f in (*.txt) do echo %%f`	loop files
`for /f "tokens=*" %%a in (f.txt) do`	read lines
`for /d %%d in (*) do`	loop folders

Goto

`:label`	define label
`goto label`	jump to label
`goto :eof`	end of script
`call :sub`	call a label

Advanced IF

Smarter decisions

Compare

`if "%a%"=="%b%"`	text equal
`if /i "%a%"=="%b%"`	ignore case
`if %n% equ 5`	equal
`if %n% neq 5`	not equal
`if %n% lss 5`	less than
`if %n% leq 5`	less or equal
`if %n% gtr 5`	greater than
`if %n% geq 5`	greater or equal

Tests

`if exist file.txt`	file exists
`if exist folder\`	folder exists
`if not exist file.txt`	file missing
`if defined name`	variable is set
`if not defined name`	variable not set
`if %errorlevel% neq 0`	last command failed

Blocks

`if x==1 (echo yes) else (echo no)`	if with else
`if "%a%"=="1" if "%b%"=="2"`	AND check
`if "%a%"=="1" goto a`	jump if true

For Loops

Repeat and process

Basic Loops

`for %%i in (a b c) do echo %%i`	loop over items
`for /l %%i in (1,1,10) do echo %%i`	count 1 to 10
`for /l %%i in (10,-1,1) do echo %%i`	count down
`for %%f in (*.txt) do echo %%f`	loop files
`for /r %%f in (*.log) do del "%%f"`	files in subfolders
`for /d %%d in (*) do echo %%d`	loop folders

File Content

`for /f %%a in (f.txt) do echo %%a`	first word of lines
`for /f "tokens=*" %%a in (f.txt) do`	whole lines
`for /f "skip=1 tokens=*" %%a in (f.txt)`	skip header line
`for /f "tokens=1,2 delims=," %%a in (f.csv)`	split by comma
`for /f %%a in ('dir /b') do echo %%a`	loop command output

Loop Tips

`%%~nxf`	file name and extension
`setlocal enabledelayedexpansion`	allow !var! in loops
`echo !count!`	read changing variable
`goto :eof`	leave a loop block

Functions & Labels

Reusable code

Labels

`:main`	define a label
`goto :main`	jump to a label
`goto :eof`	end script or function
`exit /b`	return from function
`exit /b 1`	return with error code

Functions

`call :greet Bob`	call a function
`:greet`	function starts here
`echo Hello %~1`	use the argument
`exit /b`	end of function
`call :add 2 3 result`	function with output
`set "%~3=%x%"`	return value by name

Scope

`setlocal`	start local variables
`endlocal`	end local variables
`endlocal & set x=%y%`	pass value out
`setlocal enableextensions`	turn on extensions

Structure

`@echo off`	script header
`:usage`	show help and quit
`:end`	cleanup label

Files & Folders

Common commands

Folders

`dir` `dir /s /b`	list files
`cd folder` `cd ..`	change folder
`mkdir folder`	make folder
`rmdir /s /q folder`	delete folder
`tree`	folder tree

Files

`copy a.txt b.txt`	copy file
`xcopy src dst /e /i`	copy folders
`move a.txt dst`	move file
`ren a.txt b.txt`	rename
`del a.txt`	delete file
`type a.txt`	show file text
`echo Hi > a.txt`	write file
`echo Hi >> a.txt`	append to file
`more a.txt`	page through file

Search

`find "text" a.txt`	find text
`findstr /i "text" *.txt`	search many files
`where notepad`	locate program

Text Files

Read, write, sort

Write & Read

`echo Hi > a.txt`	create or overwrite
`echo Hi >> a.txt`	append a line
`type a.txt`	show file
`type a.txt | more`	page by page
`copy con a.txt`	type text into a file
`more +5 a.txt`	skip first 5 lines

Sort & Filter

`sort a.txt`	sort lines
`sort /r a.txt`	sort in reverse
`find /c "x" a.txt`	count matching lines
`find /v "x" a.txt`	lines without text
`findstr /n "x" a.txt`	show line numbers
`findstr /r "^a.*z$" a.txt`	regex search

Combine & Compare

`copy a.txt+b.txt c.txt`	join two files
`fc a.txt b.txt`	compare files
`comp a.txt b.txt`	compare byte by byte
`certutil -hashfile a.txt MD5`	file checksum
`attrib +r a.txt`	make read-only

Errors & Exit Codes

Handle problems

Exit Codes

`echo %errorlevel%`	last exit code
`exit /b 0`	success
`exit /b 1`	failure
`cmd /c exit 5`	set a code
`if errorlevel 1 echo fail`	code 1 or higher

Safe Chaining

`cmd1 && cmd2`	run if success
`cmd1 || cmd2`	run if failed
`cmd1 & cmd2`	run both always
`(cmd1 & cmd2) || echo bad`	group commands

Handle Failures

`cmd 2>nul`	hide error messages
`cmd >nul 2>&1`	hide everything
`cmd 2>err.txt`	save errors to file
`robocopy a b /e & if %errorlevel% lss 8`	robocopy success codes
`if not exist "%f%" exit /b 1`	stop if missing

Date, Time & Random

Stamps and numbers

Date & Time

`echo %date%`	today's date
`echo %time%`	current time
`echo %time:~0,5%`	hh:mm only
`echo %date:~-4%`	year (format varies)
`wmic os get localdatetime`	stable date format
`timeout /t 10`	wait 10 seconds

Random & IDs

`echo %random%`	number 0-32767
`set /a n=%random% %% 6 + 1`	dice roll 1-6
`echo %computername%`	pc name
`echo %username%`	user name
`echo %cd%`	current folder
`echo %os%`	operating system

Useful Variables

`echo %temp%`	temp folder
`echo %userprofile%`	user home folder
`echo %appdata%`	app data folder
`echo %programfiles%`	program files folder
`echo %path%`	search path
`echo %cmdcmdline%`	how the shell started

Processes & Network

Run and monitor

Run Programs

`start notepad`	open and continue
`start /wait notepad`	open and wait
`start /min cmd`	start minimized
`runas /user:Admin cmd`	run as another user
`call script.bat`	run another script

Network

`ping -n 4 host`	ping four times
`tracert host`	trace the route
`nslookup site.com`	DNS lookup
`netstat -an`	open connections
`curl -O url`	download a file
`ipconfig /flushdns`	clear DNS cache

Services & Tasks

`sc query`	list services
`net start Spooler`	start a service
`net stop Spooler`	stop a service
`tasklist /fi "imagename eq a.exe"`	find a process
`schtasks /query`	list scheduled tasks

System & Tips

Tools and habits

System

`tasklist`	running programs
`taskkill /f /im app.exe`	stop a program
`ipconfig /all`	network info
`ping google.com`	test network
`systeminfo`	pc details
`shutdown /s /t 0`	shut down
`timeout /t 5`	wait 5 seconds

Pipes & Redirect

`cmd1 | cmd2`	pipe output
`cmd > out.txt 2>&1`	save all output
`cmd >nul`	hide output
`cmd1 && cmd2`	run if success
`cmd1 || cmd2`	run if failed

Good Habits

`Start with @echo off`	clean output
`Quote paths: "%~dp0"`	spaces safe

Keep coding. Keep learning.

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